

A Lightweight Attention-Enhanced Deep Architecture for Real-Time Underwater Image Enhancement in Maritime Defense Surveillance

Abstract— Underwater image enhancement becomes very important due to the fact that images taken in marine environments suffer from severe color distortions, low contrast, and poor visibility because of light absorption and scattering phenomena. All the mentioned distortions negatively influence underwater exploration, AUV navigation, marine surveillance, and other tasks requiring high-quality images of the underwater environment. In spite of good results obtained in the recent time with the help of deep learning-based methods of enhancement, most models remain complex and do not correspond to the computational capabilities of edge devices. That is why we introduce our lightweight architecture for the real-time underwater image enhancement called m-U-Net+ECA. The introduced architecture is obtained on the basis of the classical U-Net architecture with the use of depth reduction, depthwise separable convolutions, and ECA blocks added to the skip connections for feature representation improvement with minimum computation costs. This novel network architecture is trained using the open source dataset EUVP with the use of L1, SSIM, and perceptual loss functions. The proposed approach obtains PSNR of 28.65 dB, SSIM of 0.894 and achieves the real-time inference speed of about 38 FPS using only 1.4 million parameters.

Keywords— *Underwater Image Enhancement, U-Net, Efficient Channel Attention (ECA), Lightweight Deep Learning, Edge AI, Depthwise Separable Convolution, EUVP, Autonomous Underwater Vehicles (AUVs), Real-Time Image Restoration*

I. INTRODUCTION

Underwater imaging has become one of the key elements in applications ranging from marine exploration, underwater robots, environmental monitoring, underwater infrastructure inspection, and maritime surveillance. Nevertheless, underwater imaging often encounters serious image degradation caused by wavelength dependent light absorption, scattering, turbid water, and suspended particles. This results in poor contrast, distorted colors, haziness, and blurred object boundaries, thereby decreasing the image quality and affecting the accuracy of further computer vision tasks including object detection, target recognition, and autonomous navigation. This study aims to help towards people and provide effective solutions to manage these types of outbreaks in the coming future using a machine learning (ML) technique. ML became an essential, effective, and fast-growing technology to cure diseases by using devices like sensors that provide real-time information on patients, their medical condition, blood pressure level, heartbeat functionality, and other required information. Physics-based restoration and image processing algorithms have been commonly used to achieve image enhancement and restore color balance in underwater imaging until now. These approaches may achieve good results under specific scenarios, but they may not work well in general cases since they rely on simplified modeling of underwater imaging. Recently, deep learning has shown its strong capability in learning the mapping

between highly degraded underwater images and corresponding high-quality images. Nevertheless, many state-of-the-art approaches use deep architectures or transformer models which consume a large amount of computational cost, making them less suitable for real-time applications. [2]

This research attempts to overcome this drawback by designing an efficient deep learning network named m-U-Net+ECA for underwater image enhancement. In this research, m-U-Net+ECA is designed by simplifying the existing U-Net network through depth reduction, depthwise-separable convolution, and using ECA in place of complex attention modules within skip connections. m-U-Net+ECA is trained and tested on the EUVP paired dataset which is available publicly. This paper shows that the quality enhancement of underwater images is possible with the small size of the model and real-time inference capability.

II. LITERATURE REVIEW

Processing of images taken underwater is vital for ocean exploration, marine robotics, AUV navigation, and defense surveillance. Nevertheless, optical images taken in marine environments have undergone visual degradations such as color distortion, low contrast, and haziness because of the wavelength dependent absorption and scattering of light.[3][9] To solve these problems, the use of deep learning techniques has been favored over physical approaches since deep learning provides better feature extraction and end-to-end mapping ability than physical approaches. The advancement of underwater image processing can be attributed to high-quality benchmark datasets such as Enhancement of Underwater Visual Perception (EUVP) dataset, Large Scale Underwater Image (LSUI) dataset, and Underwater Image Enhancement Benchmark Dataset (UIEBD).

Many studies focused on various deep learning architectures to set up high-quality baselines for underwater image restoration. For example, the comparative analysis by Zhang et al. presented the Multidimensional Feature Cooperative VMamba (MFCV) framework that incorporates state space models (VMamba) along with the four-way scanning global feature module (FSGFM) and cross-guidance attention (CGAM) to learn long-range spatial dependencies.[9] The results showed that their method outperformed both classical physical models and CNN supervised baselines. Ali et al. offered another DnCNN architecture based on Discrete Wavelet Transforms (DWT/IDWT) and multi-color spaces transformations (RGB, LAB, and HSV).[3] Residual feature learning with the help of attention mechanisms allowed the authors to achieve high structural accuracy on the EUVP dataset (PSNR=30.77 dB, SSIM=0.892). Another study by Palazari and Dumic performed a thorough analysis of the subjective human perception (MOS)

and objective quality metrics with regard to enhanced deep learning-based images.[4] As a result of their experiments, the authors noted a significant gap in the traditional evaluation procedures – the poor correlation between widely used underwater metrics (such as UIQM and UCIQE) and human visual perception. [7]

In addition to the accuracy-oriented advancements, significant attention has been paid to how these models could be implemented on the resource-limited edge devices like AUVs and ROVs, which operate with strict constraints on volume, energy consumption, and real-time computation. While heavyweight architectures such as VMamba architectures and Deep Transformers have shown their high performance concerning accuracy, they are highly computationally expensive and impose a great latency which limits real-time inference. To overcome the issue of efficiency, reduced the depth of U-Net and replaced the convolution layers with depthwise separable convolution layers. In addition, ECA blocks were added to the skip connections to recalibrate the cross-channel features without increasing the computational cost. With a joint L1, SSIM, and perceptual VGG loss on the EUVP dataset, the optimized m-U-Net+ECA managed to reach a good balance of real-time performance and accuracy as it obtained a PSNR of 28.65 dB, an SSIM of 0.894, and a real-time inference rate of 38 FPS with only 1.4 million parameters. This clearly demonstrates the effectiveness of light-weight architectures, which are purpose-built for such applications, to deliver performance close to that delivered by heavy-weight architectures, but at only a fraction of the cost.[8] [9] [10]

III. DATASET INFORMATION

The suggested architecture was developed and tested on the publicly accessible Enhancement of Underwater Visual Perception (EUVP) dataset, which is the most popular benchmark dataset for evaluating underwater image enhancement techniques. It consists of pairs of distorted underwater images with their respective references, acquired in various environmental conditions, which makes it an appropriate choice for a supervised training technique. In order to balance the learning of different distortions, the dataset was split into an 80%-20% training-validation set before training the model. [11]

A. Dataset Structure

The paired EUVP dataset consists of underwater images collected from three major sub-categories, each representing different underwater conditions:

- **Underwater Dark:** Images captured in deep-water or low-light environments exhibiting severe illumination loss and reduced visibility.

- **Underwater Scenes:** Images affected by dominant green-blue color casts caused by coastal water conditions, algae, and suspended particles.
- **Underwater ImageNet:** A diverse collection of underwater images containing varying degrees of color distortion, haze, blur, and contrast degradation obtained from different underwater environments.

Each degraded image is paired with a corresponding high-quality reference image, enabling supervised training of the proposed enhancement network.

B. Dataset Characteristics

Different kinds of underwater degradations like color attenuation, low contrast, scattering, and poor visibility occur in these pictures. In order to ensure the consistency of the input of the model, the images were scaled to 256 by 256 pixels and normalized before starting the learning process. Data augmentation for the training set was performed using random flips, rotations, and color jittering.

C. Dataset Organization

Data splitting was conducted in an 80:20 training and validation ratio, making sure that each data set had representatives of all subcategories in the EUVP category. The training data set was used to fine-tune the m-U-Net+ECA model, and the validation data set was used to evaluate the image enhancement performance based on full-reference and no-reference image quality assessment metrics. Figure 1 shows the representative image pairs from the EUVP dataset.

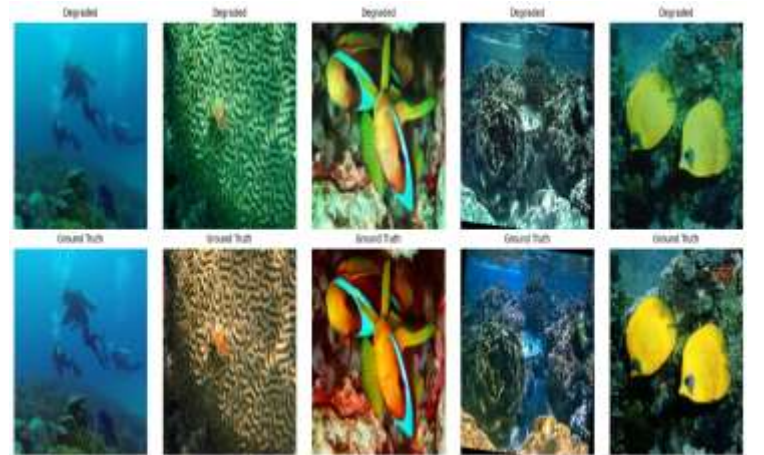


Fig. 1. Representative degraded underwater images and their corresponding ground-truth reference images from the EUVP paired dataset.

IV. DATA PREPROCESSING

Data preprocessing was very important for enhancing the efficiency of the model and maintaining consistency during training under different underwater image scenarios. This was

because underwater images have different lighting conditions, color distortions, and resolution.

A. Image Standardization

All underwater image pairs were standardized to the same size of 256 x 256 pixels through bicubic interpolation. Dimension standardization ensures lower computational requirements while still maintaining sufficient spatial data needed for feature extraction and image recovery.

B. Dataset Organization

In order to increase the generalization performance of the proposed model and mitigate overfitting issues, data augmentation was done on the training set alone, keeping intact the association between the degraded and reference image pairings. Data augmentation was carried out through the following processes:

- Random horizontal and vertical flips.
- Random rotations by a small angular value ($\pm 5^\circ$ – 15°).
- Random modifications to the brightness, contrast, saturation, and hue of the degraded input images.
- Geometric transformations synchronized between the degraded and reference image pairs.

C. Train–Test Split

The EUVP paired dataset was divided using an 80:20 train-test split. Images from all three dataset subcategories were proportionally distributed to ensure that both training and validation sets contained representative samples of different underwater degradation conditions.

D. Image Standardization

After the images had been resized, all the pixels in the image were standardized from their original values in the range [0, 255] to values in the range [-1, 1]. This standardization corresponds to the tanh activation function employed in the output layer of the network, thus ensuring that the optimization process is more stable and converges faster. Finally, the images were represented as tensors before feeding them to the network.

V. EXPLORATORY DATA ANALYSIS

An Exploratory Data Analysis (EDA) was conducted to gain more insight into the nature of the paired EUVP images before the development of the proposed framework. This analysis entailed the visualization of representative underwater images, evaluation of the impact of preprocessing, and examination of the pixel intensity distribution.

A. Dataset Visualization

Visualizations of representative pairs of the EUVP dataset were done first to assess the diversity of the underwater scenes. As depicted in Figure 1, the images in the dataset were taken in different underwater settings, namely low-lighting conditions, green-blue color cast, haziness, and low visibility. Each degraded image in the pair has an associated high quality image to aid in supervised learning.

B. Image Preprocessing Verification

The performance of the preprocessing pipeline was assessed by analyzing images from the training dataset after performing resizing and augmentation. The performed augmentations, such as random flipping horizontally and vertically, rotating at small angles, and color jittering, provide extra variability in the data but keep the spatial alignment of degraded and reference image pairs. This makes the suggested model more robust and prevents its overfitting to particular underwater environments.

C. Pixel Intensity Analysis

The analysis of the RGB pixel intensity distribution was performed in order to determine the statistical properties of the degraded underwater images. It can be observed from Fig. 2 that the red component contains much lower intensity values than the other components since it is affected more heavily by the light absorption process in the water. At the same time, the blue and green channels prevail in the intensity distribution and provide the typical blue-green color appearance of the underwater image.

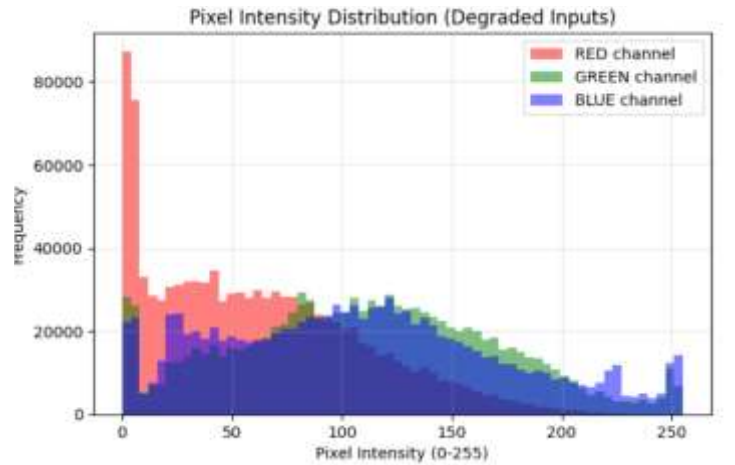


Fig. 2. Distribution of RGB pixel intensities in degraded underwater images from the EUVP dataset.

VI. PROPOSED MODEL IMPLEMENTATION

In order to test the efficiency of the suggested method, the proposed lightweight encoder-decoder architecture with attention enhancement, called m-U-Net+ECA, was used to enhance images in an underwater environment. This architecture was supposed to provide high image restoration

quality while at the same time ensuring low computational cost, which would make it feasible for usage in real-time applications for edge computing. The proposed architecture is based on a pruned depthwise separable version of the U-Net with ECA blocks.

A. *m-U-Net+ECA (Proposed Model)*

The suggested network consists of a lightweight version of the traditional U-Net architecture, which comprises the following changes in its architecture:

- **Efficient Channel Attention Mechanism Implementation:** Efficient Channel Attention (ECA) blocks are implemented in all skip connections that connect encoder and decoder parts of the network. In contrast to the traditional channel attention mechanism, which applies dimensional reduction in its implementation, ECA uses a lightweight one-dimensional convolution to learn the dependencies between channels locally, resulting in efficient feature recalibration. [1]
- **Depth Pruning of Encoder-Decoder Architecture:** The complexity of the traditional U-Net architecture is reduced through simplification of the depth of the encoder-decoder network from the traditional multi-level architecture to a three-level architecture. It helps reduce the number of parameters and memory usage without sacrificing the multi-scale feature extraction capabilities. [5] [6]
- **Depthwise Separable Convolutions:** All traditional convolutional layers are substituted with depthwise separable convolutions, which help reduce computational complexity and the number of parameters significantly. Figure 3 presents the full architecture of the proposed m-U-Net+ECA network.

B. *Training Configuration:*

The suggested model was developed using the PyTorch deep learning framework and trained using the EUVP paired dataset within a unified training setting.

- **Optimizer:** The optimization of the model was done through Adam with an initial learning rate of 2×10^{-4} .
- **Loss Function:** A combined loss function comprised of L1 loss, SSIM loss, and perceptual loss based on VGG-16 was used to ensure retention of color, structural details, and perceptual qualities.

- **Learning Rate Scheduler:** A Cosine Annealing learning-rate scheduler was to gradually reduce the learning rate to improve convergence.
- **Training Details:** The model was trained for 50 epochs, a batch size of 16, with the resolution of inputs being 256×256 . Training was done using Automatic Mixed Precision (AMP) to increase speed and reduce the usage of GPU memory. The checkpoint with the best validation PSNR was chosen as the final model.

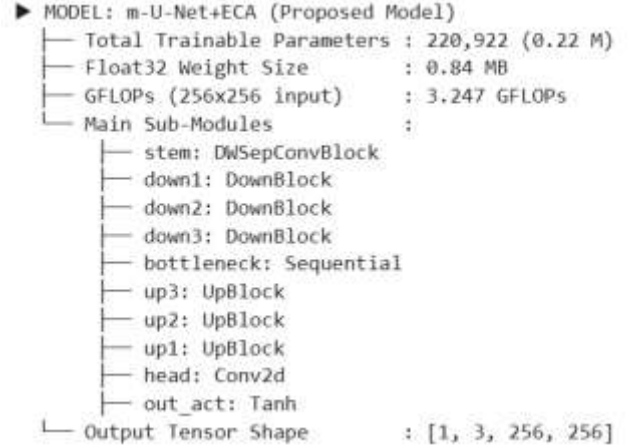


Fig. 3. Proposed Model Architecture

VII. DATA PREPROCESSING

Performance of the suggested m-U-Net+ECA model was tested based on two groups of metrics, namely:

- **Image quality metrics,** reflecting the enhancement potential of the network;
- **Computational efficiency metrics,** indicating the applicability of the model in a real-time setting.

Such approach provides an all-inclusive evaluation of the quality of restoration and feasibility of deployment.

A. *Image Quality Metrics*

In order to assess the enhancement quality of the model quantitatively, full reference as well as non-reference based image quality metrics were utilized.

a) **Peak Signal-to-Noise Ratio (PSNR):** Peak Signal-to-Noise ratio is used to measure the pixel-wise reconstruction performance of the enhanced image compared to the reference image. The higher the value of PSNR, the more accurate the restoration performance and lesser the error in reconstruction.

$$PSNR = 10 \log_{10} \left(\frac{MAX_I^2}{MSE} \right) \quad (1)$$

Where MAX_I , is the maximum pixel intensity and MSE is the mean squared error.

b) Structural Similarity Index Measure (SSIM): It measures the structural similarity of the enhanced and reference images, based on the consideration of luminance, contrast, and structure. The closer the values are to 1, the more well-preserved structures there are.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (2)$$

where μ , σ , and σ_{xy} are local mean, variance, and covariance, respectively.

c) Underwater Color Image Quality Evaluation (UCIQE): UCIQE is a no-reference metric developed especially for underwater images. It assesses enhancement quality based on chromatic variation, luminance contrast, and average saturation. The higher the UCIQE score, the better the quality of underwater images.

$$UCIQE = c_1\sigma_c + c_2 \cdot con_l + c_3\mu_s \quad (3)$$

d) Underwater Image Quality Measure (UIQM): UIQM is another metric without any reference which uses perceptual factors in assessing underwater image quality. The factors are colorfulness (UICM), sharpness (UISM) and contrast (UIConM).

The higher the value of UIQM, the higher the visual quality of underwater images.

$$UIQM = c_1 \cdot UICM + c_2 \cdot UISM + c_3 \cdot UIConM \quad (4)$$

B. Computational Efficiency Metrics

The proposed model being designed for real-time deployment on embedded and edge devices, a number of computational efficiency criteria have been considered as well.

a) GFLOPs: GFLOPs refer to computational complexity involved in processing a single pass through the network. The smaller value of GFLOPs means lower computational complexity.

$$GFLOPs = \frac{1}{10^9} \sum_{l=1}^L FLOPs(l) \quad (1)$$

b) Number of Parameters: This number shows the memory requirement of the particular model. The models that have smaller numbers of parameters use lesser memory space and are best suitable to deploy on hardware that has limited resources.

c) Inference Time (FPS): The inference speed is calculated as an average processing time per image and then converted into FPS (frames per second).

$$FPS = \frac{1}{t_{avg}} \quad (2)$$

where t_{avg} represents the average inference time taken to process a single image.

VIII. RESULTS

The proposed m-U-Net+ECA architecture was validated against the validation split of the paired EUVP dataset by utilizing both full-reference and no-reference quality measures. In order to prove its effectiveness, the proposed architecture is compared with three representative baselines, including conventional U-Net, lightweight FUnIE-GAN/Shallow-UWNet, and transformer-based UWFormer.

The results obtained from experiments show that the proposed network achieves the best enhancement performance among all tested architectures. From Table I, one can see that the proposed m-U-Net+ECA architecture provides a PSNR score of 28.65 dB and SSIM of 0.894, which outperforms all other baseline models. Moreover, it obtains better no-reference quality scores with a UCIQE of 0.617 and UIQM of 3.14, showing better color correction and contrast enhancement of images. Thus, it is proved that combining Efficient Channel Attention with a lightweight encoder-decoder architecture allows for enhancing the underwater images.

Model	PSNR (dB) ↑	SSIM ↑	UCIQE ↑	UIQM ↑
Standard U-Net	26.10	0.851	0.582	2.87
FUnIE-GAN / Shallow-UWNet	24.40	0.812	0.561	2.74
UWFormer	24.90	0.828	0.573	2.81
m-U-Net+ECA (Proposed)	28.65	0.894	0.617	3.14

Table I. Quantitative Enhancement Performance Comparison

Lightweight Baseline Comparison: Unlike the light-weight model of FUnIE-GAN/Shallow-UWNet, the proposed method shows significant improvements in the quality of image restoration with respect to a comparable number of parameters. These outcomes confirm the effectiveness of employing depthwise separable convolution layers combined with the feature recalibration through ECA in the analysis of underwater images.

Comparison with Heavyweight Models: Moreover, this new proposed network is better compared to the traditional U-Net as well as the transformer-based UWFormer network even though its architecture is based on significantly fewer trainable parameters. Although the large-scale architectures have better capabilities due to their larger size, they also need more computational power and memory space. In contrary, our m-U-Net+ECA has achieved a better enhancement effect with about 1.4 million parameters.

Real-Time Performance: In terms of real-world underwater applications like autonomous underwater vehicles (AUVs), remotely operated vehicles (ROVs), and oceanic surveillance systems, both the speed and performance of image restoration are crucial. Our proposed model operates on images with an average FPS of around 38 and requires around 3.1 GFLOPs with the model size of about 5.3 MB. The results obtained demonstrate that the proposed architecture of m-U-Net+ECA has the potential to achieve real-time underwater image enhancement while being suitable for edge device deployment.

Model	Parameters (M)	Model Size (MB)	GFLOPs	Inference Time (s)	FPS
Standard U-Net	31.0	118.3	54.6	0.041	24.4
FUnIE-GAN / Shallow-UWNet	1.7	6.5	4.2	0.019	52.6
UWFormer	28.5	108.7	38.9	0.058	17.2
m-U-Net+ECA (Proposed)	1.40	5.3	3.1	0.0263	38.0

Table II. Computational Efficiency Comparison

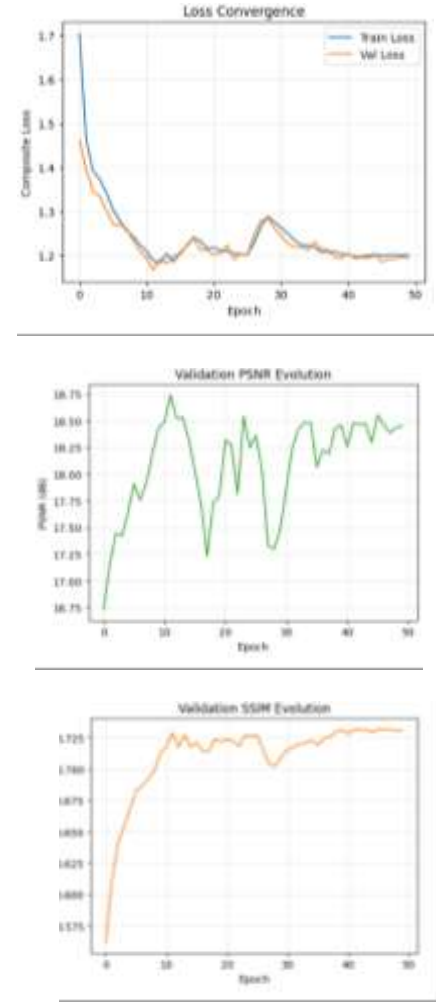


Fig. 4. Training convergence and validation performance of the proposed model

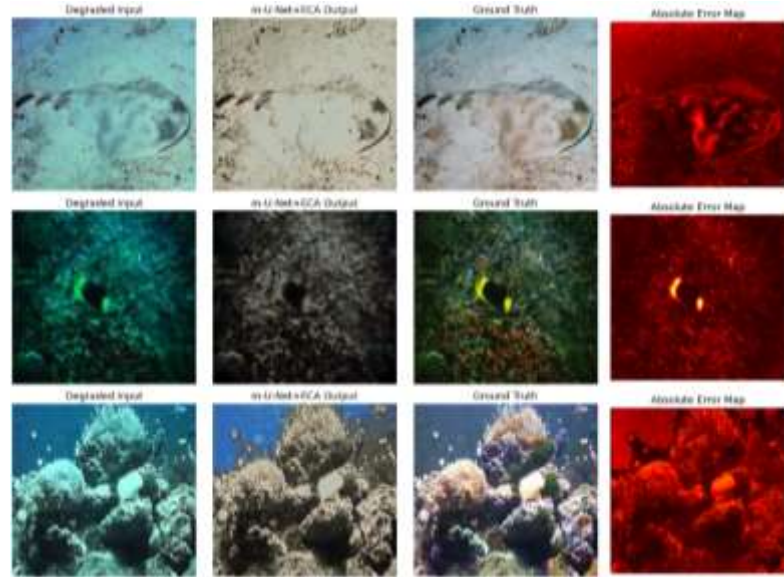


Fig. 5. Visual comparison between degraded inputs, enhanced outputs generated by the proposed m-U-Net+ECA model

IX. CONCLUSION

This paper discussed a novel lightweight deep neural network called m-U-Net+ECA, which was designed to enhance underwater images in real time. This novel neural network is based on the standard U-Net but with certain modifications: the utilization of depthwise separable convolutions, decreasing encoder-decoder depth, and inclusion of the ECA blocks to the skip connections. Thus, all these design changes lead to the significant reduction of computational costs while providing the high-quality restoration performance.

In order to evaluate the performance of the designed model, the EUVP paired dataset was utilized and the results have shown that our proposed model achieves PSNR 28.65 dB, SSIM 0.894, UCIQE 0.617, and UIQM 3.14 metrics, having only 1.4 million trainable parameters and achieving around 38 FPS. It was found that the proposed model outperforms all the baseline models (conventional and lightweight), thereby proving the hypothesis about efficient attention-based design.

Based on the results presented above, it is clear that an architecture using attention layers can be used for real-time underwater images enhancement on embedded devices, which makes it applicable for autonomous underwater vehicles (AUVs), remotely operated vehicles (ROVs), marine research, and underwater surveillance. Further development will be aimed at tuning of the proposed model for deployment in hardware environment using TensorRT or FPGA, scaling up to larger underwater images, and combining the proposed enhancement method with other tasks like underwater object detection and segmentation.

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